Sunghyun Park

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Research Interest

Robot Learning, Optimal Control, Whole-Body Control, Legged Locomotion, Cross-Embodiment Generalization

Education

Yonsei University, *Seoul, Republic of Korea* Mar 2021 – Present

B.E. in Mechanical Engineering

- Expected Graduation: Feb 2027
- Cumulative GPA: 4.11/4.3 (3.92/4.0)
- Relevant Courses: Reinforcement Learning, Mechatronics, Mechanical System Control, Engineering Numerical Analysis, Applied Dynamics
- 18-month absence for mandatory military service as a KATUSA (Korean Augmentation to the United States Army) medic

Publications & Working Papers

- H. Park*, J. Chae*, M. Park*, **S. Park**, H. Yoo, S. Choi, S. Yoo, J. Seo, S. Idrees, J.-S. Hyun, J. Lee, R. Horowitz, Y. Lee, and J. Choi, "CoHuB: A Simulation Benchmark for Multi-Humanoid Collaboration", Accepted to the **NeurIPS 2026 Workshop on Robot Learning**; *under review* at **ICLR 2027**. (Link) *Equal contribution.

- **S. Park** and H. Kim, "JOSE: Reconstructing Privileged State from Joint History for Humanoid Motion Control", submitted to **2027 IEEE International Conference on Robotics and Automation (ICRA)**, *under review* (Link)

Research Experience

Machine Learning and Control Systems (MLCS) Lab, Yonsei University Nov 2025 – Present

Undergraduate Researcher, *Adviser*: Prof. Jongeun Choi

VLA Model Training on RB-Y1

- Implemented a Meta Quest-based teleoperation pipeline for RB-Y1 (a wheeled dual-arm manipulator)
- Collected demonstrations and trained VLA models (GR00T, $\pi_{0.5}$, X-VLA) on RB-Y1

Multi-Humanoid Benchmark

- Proposed CoHuB, a collaborative multi-humanoid benchmark of 10 tasks (8 two-robot tasks, 2 three-robot tasks)
- Implemented benchmark tasks and trained multi-agent baseline policies

Dynamic Robotic Systems (DYROS) Lab, Seoul National University Jul 2025 – Sep 2025

Undergraduate Intern, *Adviser*: Prof. Jaeheung Park

Whole-Body Motion Imitation on TOCABI

- Retargeted human motion to TOCABI, a full-size torque-controlled humanoid
- Trained whole-body motion-tracking policies with goal-conditioned RL

Selected Projects

UMMA: University Mobile Management Agent Mar 2026 – Jun 2026

- Creative Product Design, Department of Mechanical Engineering, Yonsei University, 2026 Spring
- Built the robot's control and task-execution stack in ROS2 for an autonomous building management system
- Keywords: ROS2, YOLO, SLAM, Hardware Design

SOLO: State Estimation with Only Leg Observation Dec 2025 – Apr 2026

- Estimated base state from leg observation history and deployed it on **DEXTRA**
- Keywords: Locomotion, State Estimation

PALMA: A Robotic Hand Project Dec 2025 – Feb 2026

- Fabricated every part in-house by 3D printing and assembled a fully tendon-driven hand
- Vision-based teleoperation
- Keywords: Computer Vision, Tendon Actuation

DEXTRA: A Low-Cost Bipedal Robot Aug 2025 – Jan 2026

- Designed, printed, and assembled a 12-DoF bipedal robot from scratch for under $800 in parts
- `DeepMimic`-based bipedal walking motion learning
- Keywords: Motion Retargeting, RL

Selected Awards & Honors

Grand Prize, Course Outcome Idea Competition, Yonsei University 2026 Summer

- UMMA: University Mobile Management Agent
- Qualified for the National Championship

Academic Excellence Scholarship, Yonsei University 2022 Spring – Present

- Ranked 1st out of 231 students in the Department of Mechanical Engineering (Fall 2024)
- Selected for 6 semesters, excluding period of military service

National Scholarship (Type I & II), Korea Student Aid Foundation (KOSAF) 2021 Fall – Present

- Merit-based financial support awarded for maintaining excellence in undergraduate studies
- Selected for 7 semesters, excluding period of military service

3rd Place, 2025 STEAMCUP IRC TurtleBot3 AutoRace Nov 2025

- Implemented autonomous driving framework for TurtleBot3
- Built on: ROS2, Ubuntu Linux, FlexBE, Raspberry Pi

Extracurricular Activities and Leadership

YAI, Artificial Intelligence Club, Yonsei University Jun 2024 – Present

- Studied machine learning fundamentals and reviewed recent robot learning literature through paper discussions
- AI Projects: Learning bipedal walking for DEXTRA, RL-HFT project

RoboIn, Robotics Club, Yonsei University Mar 2024 – Present

- Executive Staff (2024–2025), Vice President (2026)
- Robot Projects: DEXTRA (bipedal robot), PALMA (robotic hand), TurtleBot, Manipulator
- Conducted Seminars: Basics of Humanoids; Robot Learning: Starting from Reinforcement Learning

Skills

Programming Languages	Python, C, C++, MATLAB
Frameworks/Tools	ROS2, Docker, Isaac Gym, Isaac Lab, MuJoCo, Fusion 360, Inventor, Creo, ANSYS
Hardware	Raspberry Pi, myRIO, Arduino, Jetson series, Dynamixel series, TurtleBot
Languages	Korean (Native), English (Fluent, TOEFL iBT 5.5/6)